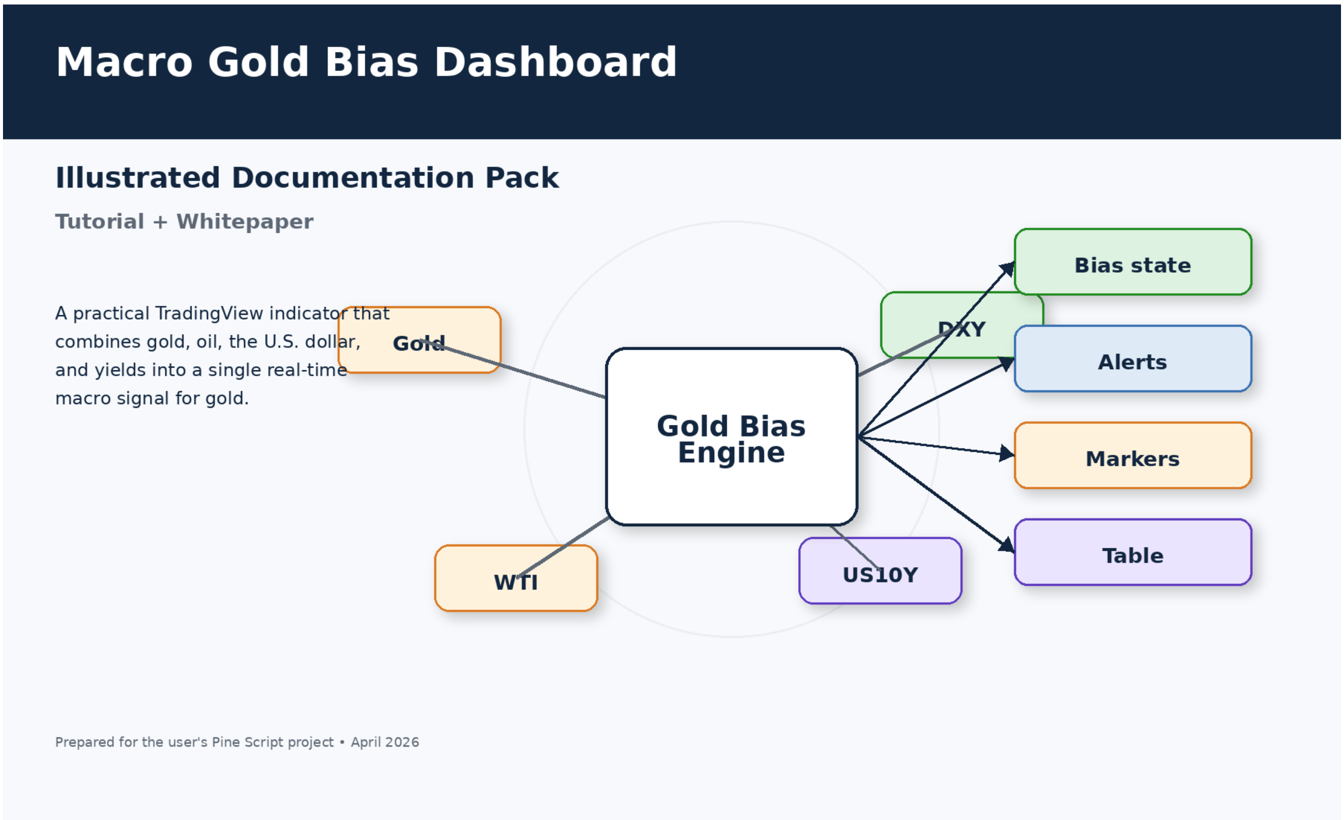

Macro Gold Bias Dashboard — Illustrated Tutorial

How to install, configure, interpret, and tune the TradingView Pine indicator

Prepared from the current Pine Script revision, including the typed oilScore fix.

Contents

1. Quick start

This indicator is built to answer a simple practical question: is the current macro backdrop supportive or hostile for gold? Instead of watching gold, oil, the U.S. dollar, and yields separately, the script converts them into one interpretable bias state.

For the cleanest first run, add the indicator to an XAUUSD chart, keep the default parameters, and use a 1-hour, 4-hour, or daily timeframe. The indicator works on any chart, but those three timeframes usually make the macro relationships easiest to read.

- Open TradingView and launch the Pine Editor.
- Paste the script into a new indicator tab.
- Confirm that the oilScore declaration uses an explicit type: float oilScore = na.
- Save the script and add it to the chart.
- Check the four symbol inputs and remap them if your feed uses different tickers.
- Wait until enough bars are loaded so the EMA, ROC, and rebased lines can initialize.

```
float oilScore = na
if dataOk
    oilScore := 0.0
    if oilState == -1 and macroTailwind >= 1
        oilScore := 0.75
    else if oilState == -1 and macroTailwind <= -1
        oilScore := -0.75
```

That explicit float declaration matters because Pine cannot infer a numeric type from bare na during variable creation. Once the variable is typed, the subsequent := assignments work as intended.

2. Market inputs and why they matter

The model uses four inputs, but they do not all play the same role. Gold is the confirmation leg. DXY and yields are the main macro headwind or tailwind legs. Oil is a context leg that nudges the score rather than dominating it.

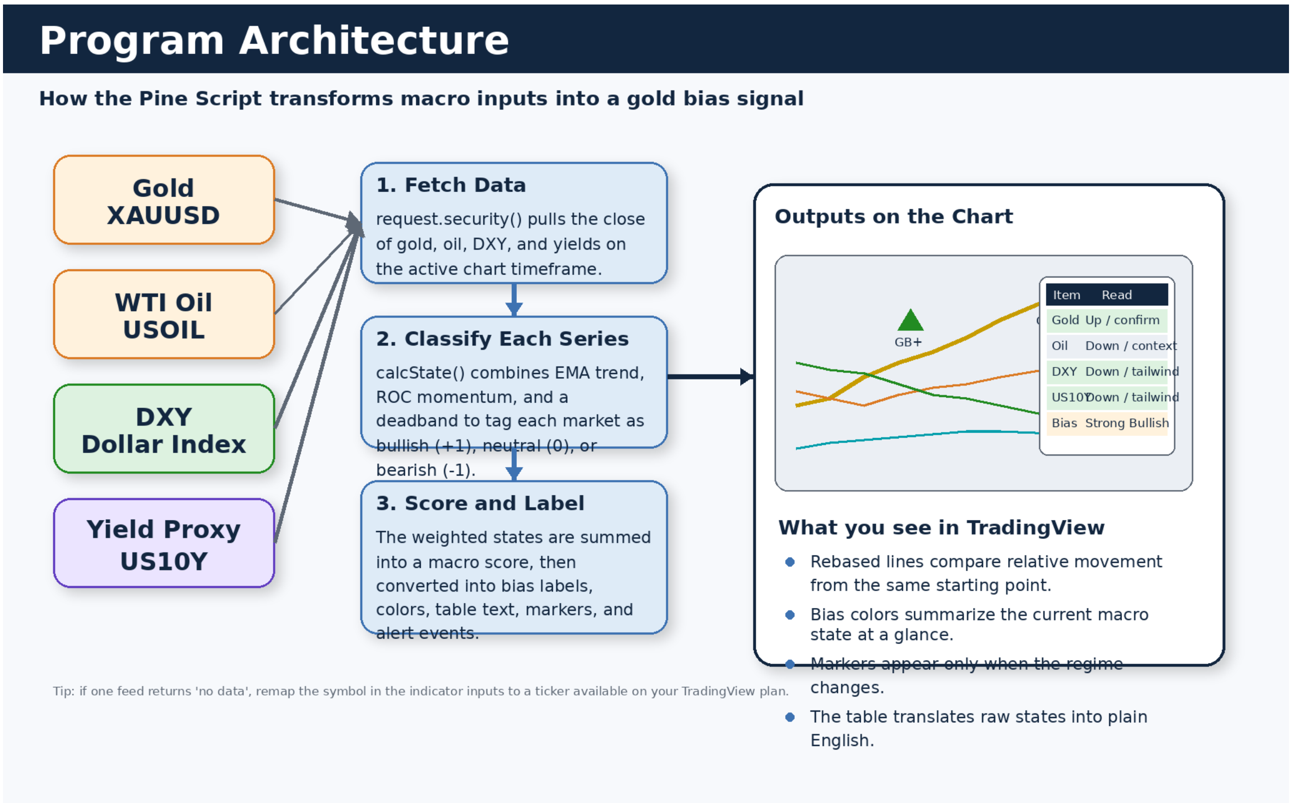

Illustration 1. End-to-end program architecture: four market inputs become states, a weighted score, and chart outputs.

Input	Role in the model	Typical meaning when rising	Weighting logic
Gold (XAUUSD)	Confirms whether gold itself is trending with momentum	Supports a bullish gold reading	+1.00 × goldState
WTI oil	Adds inflation/growth context	Can help or hurt depending on DXY and yields	Context modifier only
DXY	Primary inverse macro driver	Usually a headwind for gold	-1.25 × dxyState
Yield proxy	Primary inverse macro driver	Usually a headwind for gold	-1.25 × yldState

A useful mental model is this: gold confirms, the dollar and yields drive, and oil contextualizes. If you remember only one thing, remember that the script deliberately treats DXY and yields as more important than oil.

3. The state engine

Each input series is converted into a simple state: bullish (+1), neutral (0), or bearish (-1). The script does this with the calcState() helper, which combines a trend filter and a momentum filter.

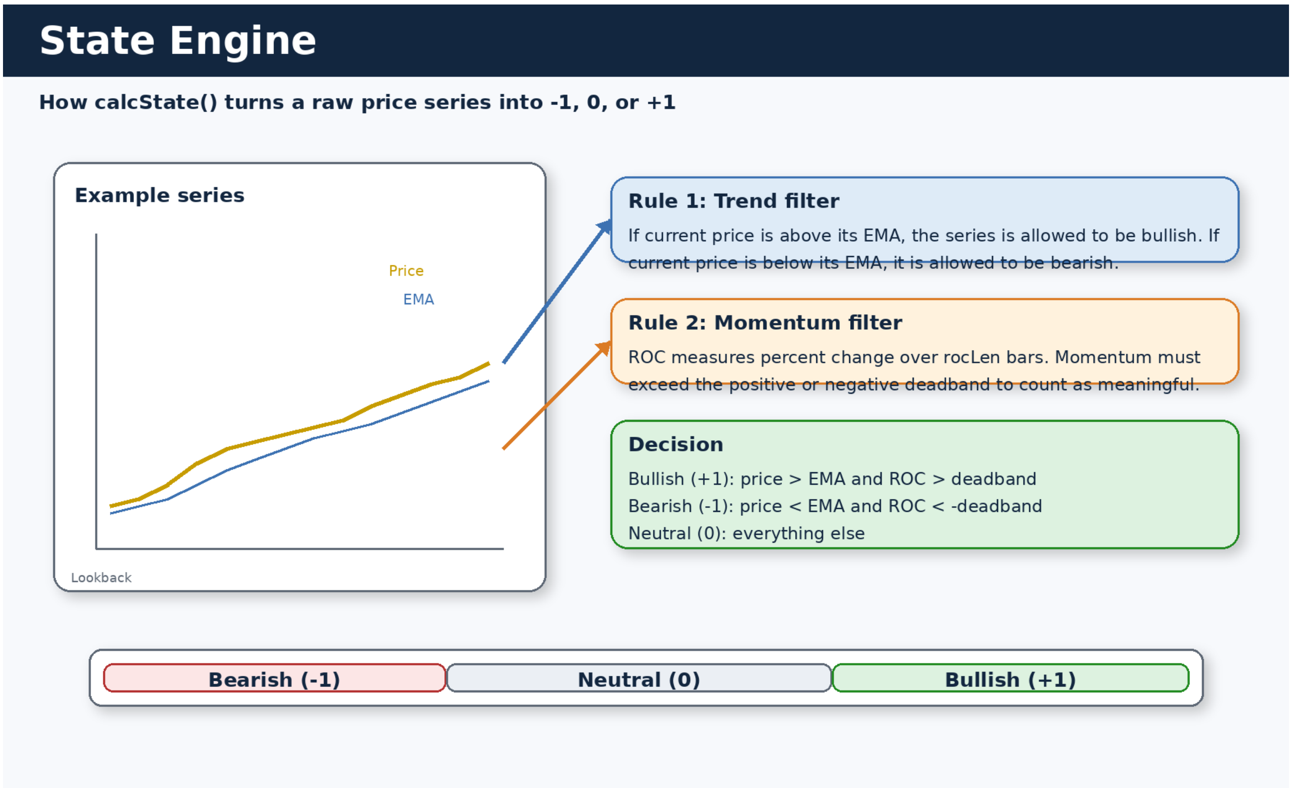

Illustration 2. The state engine uses price versus EMA and ROC versus a deadband to produce -1, 0, or +1.

```
calcState(s, emaLength, rocLength, deadband) =>
    ema_ = ta.ema(s, emaLength)
    roc_ = na(s[rocLength]) or s[rocLength] == 0 ? na : 100.0 * (s / s[rocLength] - 1.0)
    st_ = na(ema_) or na(roc_) ? 0 : s > ema_ and roc_ > deadband ? 1 : s < ema_ and roc_ < -deadband
    [st_, ema_, roc_]
```

- Bullish state: price is above its EMA and momentum is meaningfully positive.
- Bearish state: price is below its EMA and momentum is meaningfully negative.
- Neutral state: either the trend and momentum disagree, or momentum is too small to trust.

The deadband is important because it suppresses tiny momentum fluctuations. Without it, the indicator would flip states too easily whenever price drifted sideways.

4. The scoring model

Once the four series are classified, the script turns those states into a single score. This is the heart of the program. The score is not a forecast of exact returns; it is a compact measure of how supportive the macro mix is for gold right now.

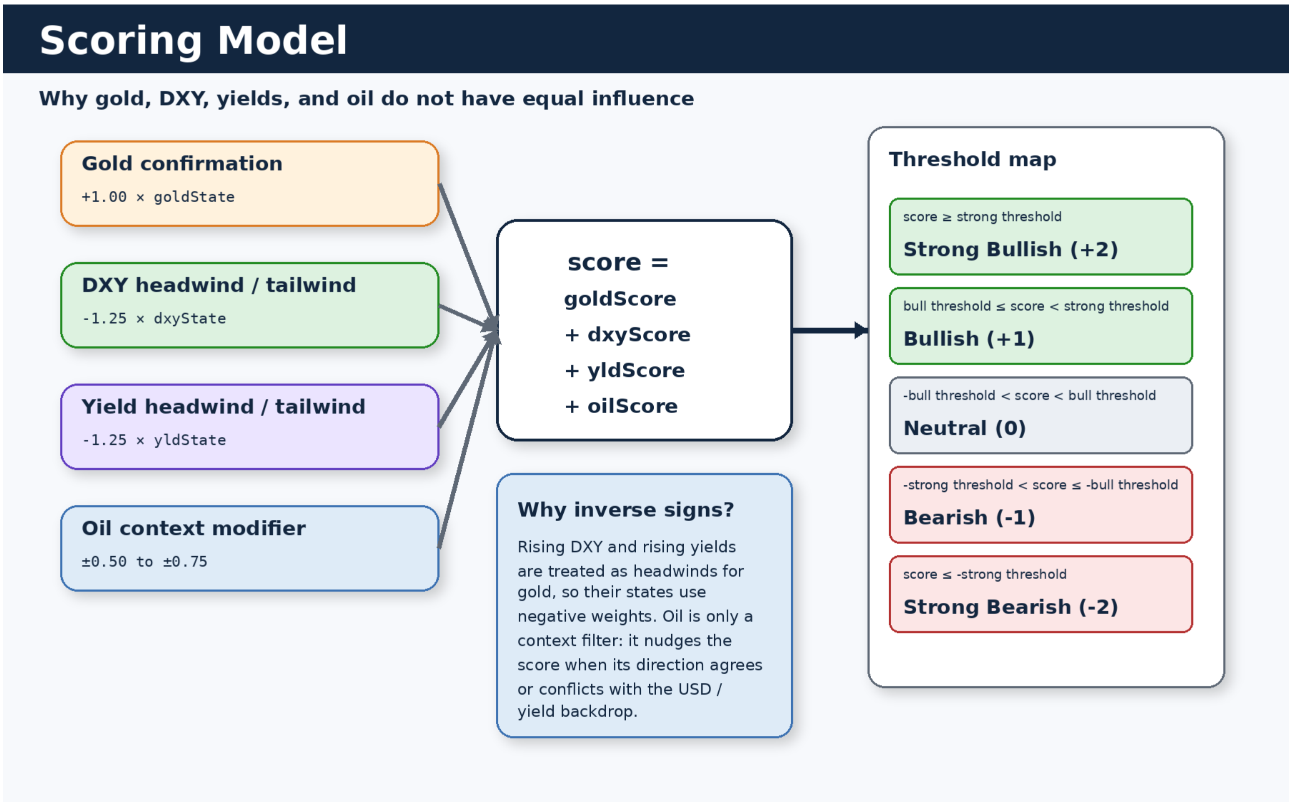

Illustration 3. Weighted state inputs, threshold mapping, and the reason DXY and yields use inverse signs.

```
score = goldScore + dxyScore + yldScore + oilScore
```

Component	Formula	Interpretation
Gold confirmation	goldState × 1.00	Gold's own trend and momentum confirm the backdrop
DXY effect	-dxyState × 1.25	A rising dollar is treated as a gold headwind
Yield effect	-yldState × 1.25	Rising yields are treated as a gold headwind
Oil context	±0.50 to ±0.75	Oil only nudges the score when it agrees or conflicts with the macro backdrop

The oil modifier uses macroTailwind = (-dxyState) + (-yldState). If oil falls while DXY and yields are also falling, the script treats that as gold-friendly. If oil falls while DXY and yields are rising, it treats the move as a disinflation headwind for gold instead.

- score ≥ strong threshold → Strong Bullish
- bull threshold ≤ score < strong threshold → Bullish
- -bull threshold < score < bull threshold → Neutral
- -strong threshold < score ≤ -bull threshold → Bearish
- score ≤ -strong threshold → Strong Bearish

5. Reading the chart

The indicator gives you several layers of output at once: rebased lines for visual comparison, background tint for instant regime recognition, chart markers for regime changes, a floating table for plain-English interpretation, and alerts for automation.

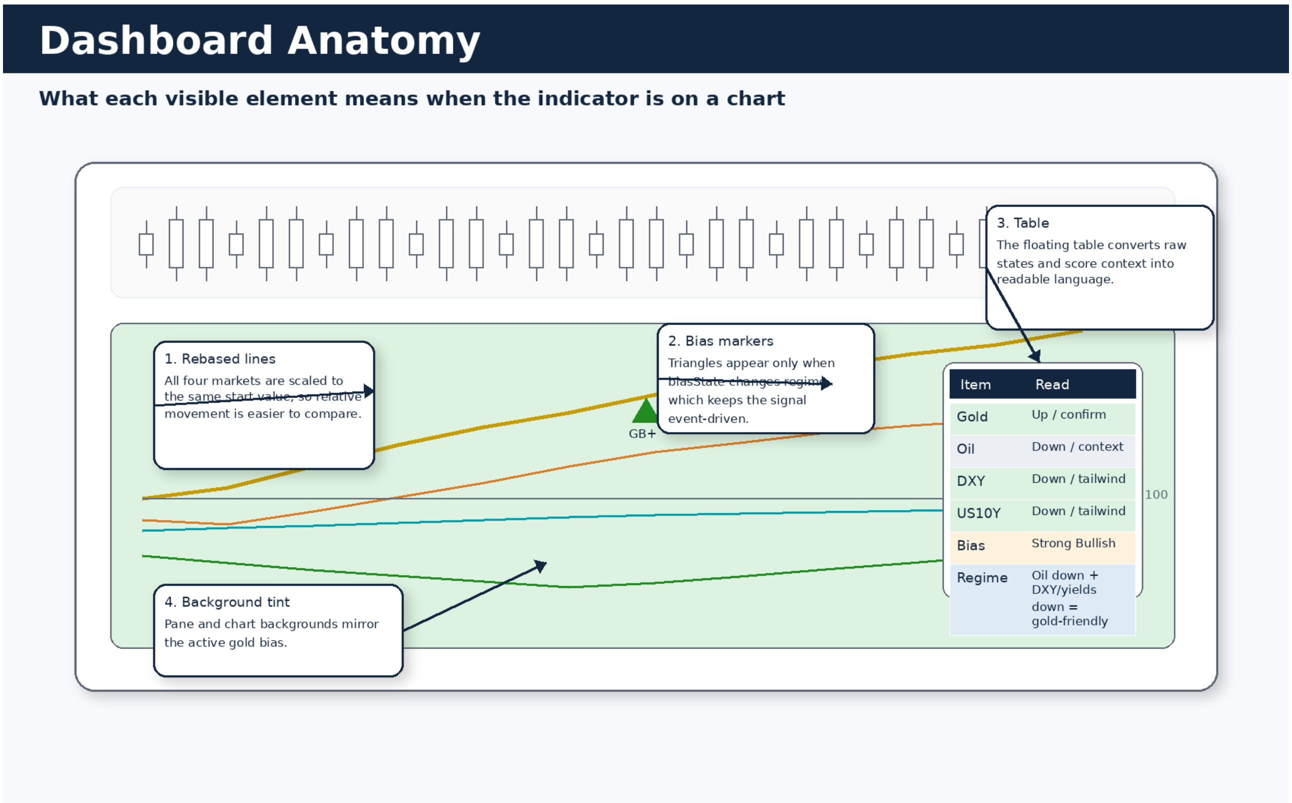

Illustration 4. A labeled mock-up of the indicator pane and its main visual elements.

- Rebased lines answer: which market has moved the most since the selected lookback point?
- Background tint answers: what is the active gold bias right now?
- Markers answer: when did that bias change?
- The table answers: why is the model saying this?

Because the comparison lines are rebased, they are not showing raw price levels. They are showing relative movement from a common starting point. That makes cross-asset behavior much easier to compare visually.

6. Scenario playbook

This section turns the logic into a quick-reading playbook. The same oil move can mean different things for gold depending on what DXY and yields are doing. That is why the script does not hard-code a simple 'oil down equals gold up' rule.

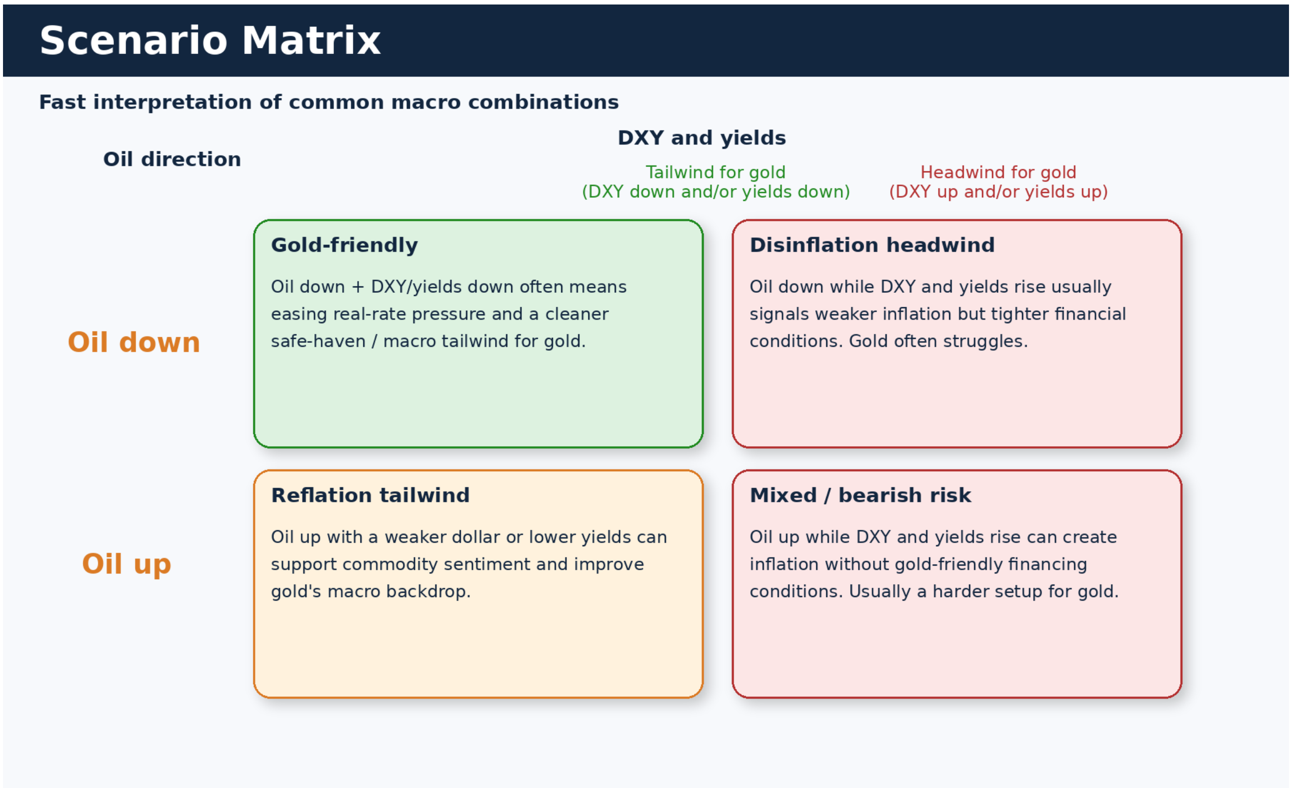

Illustration 5. A simple scenario matrix for interpreting oil alongside DXY and yields.

- Oil down + DXY down / yields down: usually the most gold-friendly combination.
- Oil down + DXY up / yields up: often a harder setup for gold because tighter financial conditions dominate.
- Oil up + DXY down / yields down: can support reflation and commodity sentiment.
- Oil up + DXY up / yields up: often mixed or bearish for gold because financing conditions are unfriendly.

In practice, many users find it helpful to read the table from right to left: first the Bias row, then DXY and US10Y, and only then Oil. That sequencing mirrors the script's weight hierarchy.

7. Alerts and signal changes

The indicator is event-driven. It does not fire alerts continuously inside a regime. Instead, it emits alerts only when the bias changes from one regime to another. That design keeps noise down and makes the alerts more actionable.

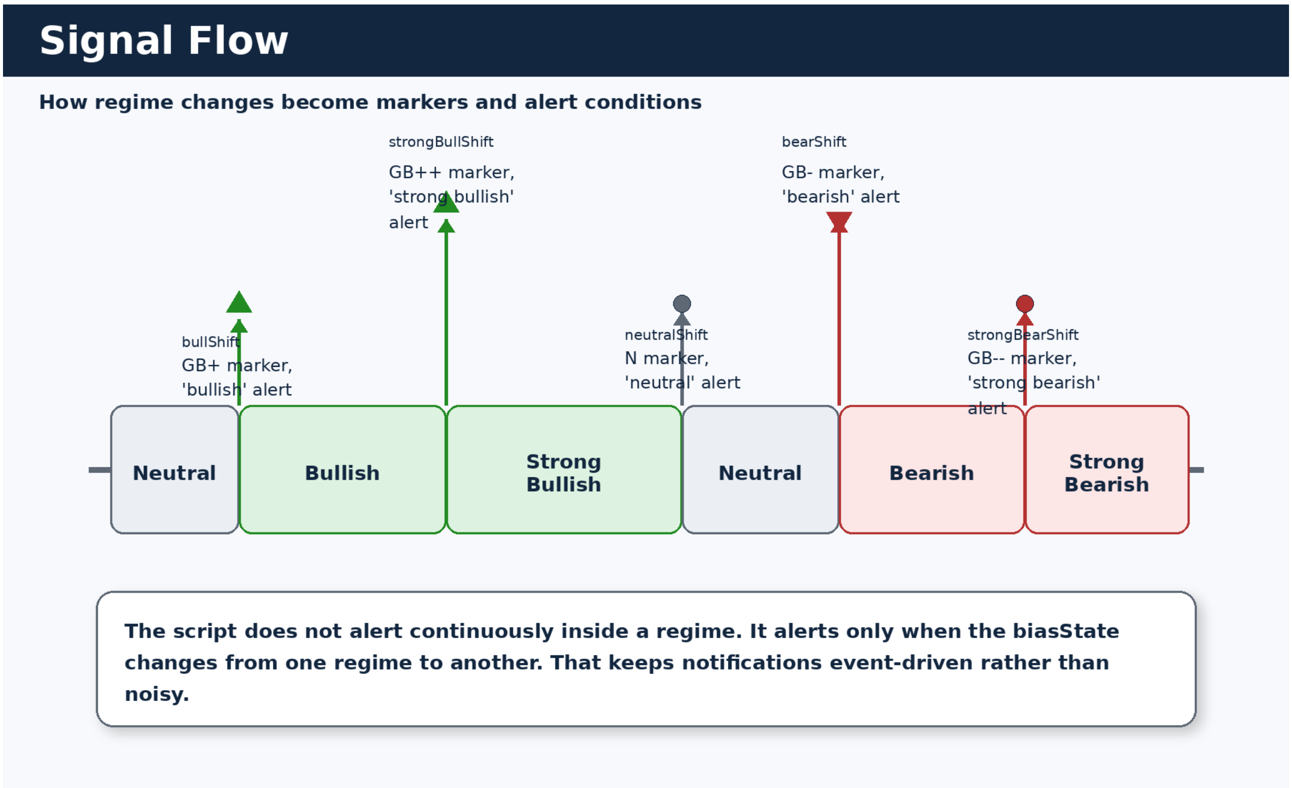

Illustration 6. The five alert conditions and the regime transitions that trigger them.

Condition	When it triggers	What it means
bullShift	biasState turns +1 from 0 or a bearish state	The setup has improved into a bullish regime
strongBullShift	biasState turns +2 from anything below +2	The setup has strengthened further
neutralShift	biasState returns to 0 from any non-neutral state	The edge has faded or become mixed
bearShift	biasState turns -1 from 0 or a bullish state	The backdrop has deteriorated
strongBearShift	biasState turns -2 from anything above -2	The bearish setup has intensified

If you want fewer alerts, raise the thresholds or increase the deadband. If you want earlier alerts, reduce those settings cautiously and accept that the script will become more sensitive.

8. Parameter tuning

The default settings are a good starting point, but the best values depend on your timeframe and how early versus how reliable you want the signal to be.

Use case	EMA length	ROC lookback	Deadband	Comments
1-hour chart	21–34	3–5	0.15–0.25	More responsive, but can whipsaw during range-bound sessions
4-hour chart	34–55	5–8	0.20–0.35	A balanced macro swing-trading profile
Daily chart	34–55	5–10	0.25–0.40	Slower but cleaner regime changes

- Lower EMA length = faster trend detection, more noise.
- Lower ROC lookback = faster momentum detection, more noise.
- Lower deadband = more state changes.

- Lower thresholds = more bullish and bearish labels.
- Higher rebase length = longer visual comparison window.

A simple workflow is to start with the defaults, observe how often the bias flips on your preferred chart, then change only one parameter at a time. That keeps the tuning process interpretable.

9. Troubleshooting

Symptom	Likely cause	What to do
The table says 'Check symbol inputs'	One or more symbols returned no data	Open the indicator settings and remap the missing feed
score is na	Not enough bars for lookbacks	Load more chart history or reduce lookback values
The indicator flips too often	Settings are too sensitive for the timeframe	Raise deadband or thresholds; consider a longer EMA
Signals feel late	Trend and momentum filters are too slow	Lower EMA and ROC lengths modestly
Oil behavior feels confusing	Oil is a context filter, not the primary driver	Read DXY and yields first, then use oil as confirmation
Compiler error about na typing	A variable was declared with bare na	Use an explicit type such as float oilScore = na

Feed differences matter. If your account does not have the default ticker source, the logic can still work well after you remap the symbols to an available gold, oil, dollar, and yield feed.

10. Operating guidelines

This indicator is best used as a context layer, not as a standalone entry system. It tells you whether the cross-asset backdrop is helping or hurting gold, but it does not replace price structure, support and resistance, event risk, or position sizing.

- Use the bias to filter trades, not to force them.
- Confirm with the actual gold chart before acting.
- Treat strong bullish and strong bearish states as macro winds, not guarantees.
- Be especially careful around major macro releases when correlations can temporarily break.

A strong workflow is to combine the bias with your existing execution method. For example, use the script to define directional preference, then wait for your own price-action trigger on gold itself.

The companion whitepaper in this package explains the design choices behind the model and outlines ways to extend it into a strategy version.